Exploration and visualisation of metabolic networks for model organisms

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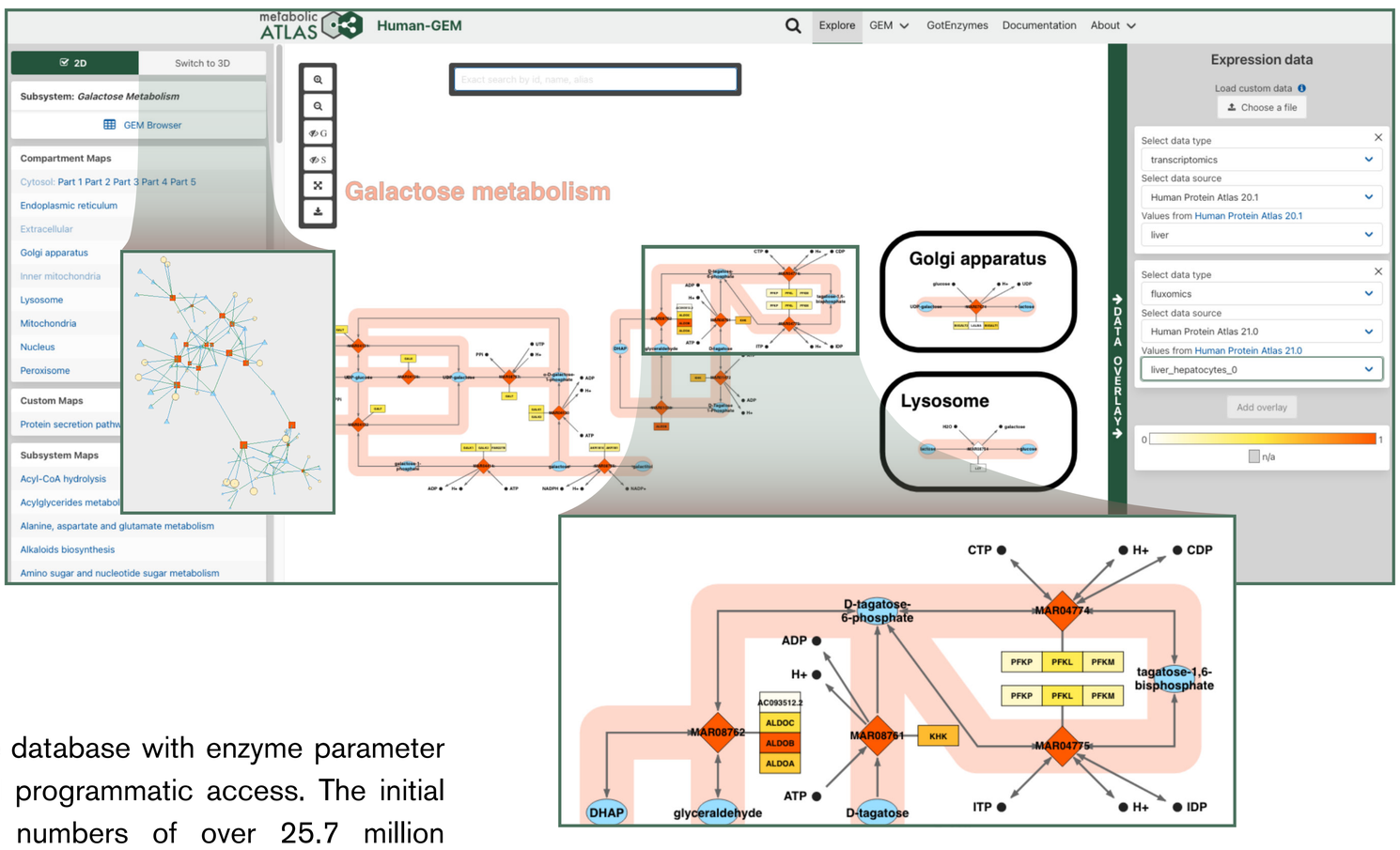
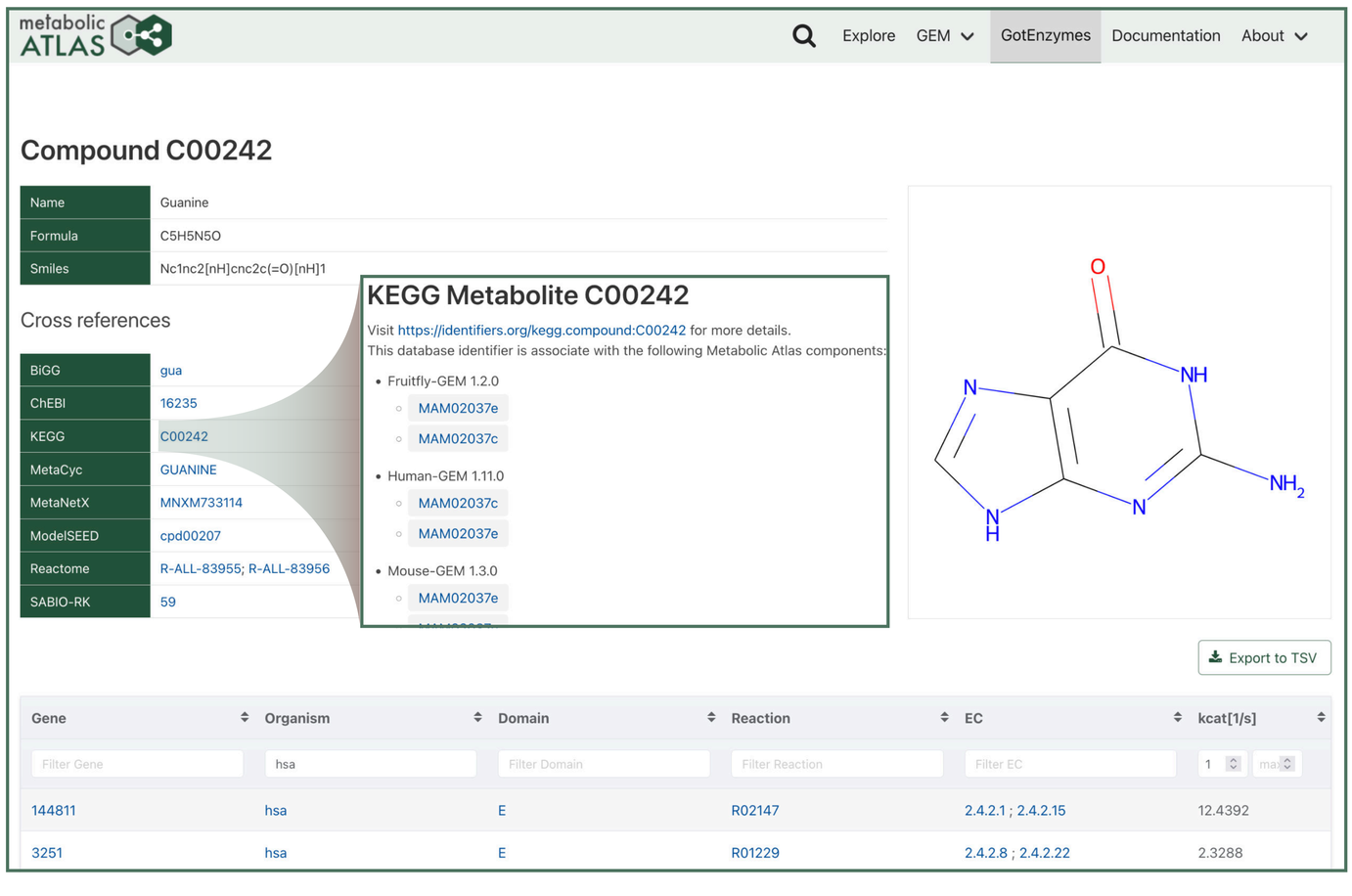
Metabolic Atlas 3 has been publicly released and is freely available at https://metabolicatlas.org

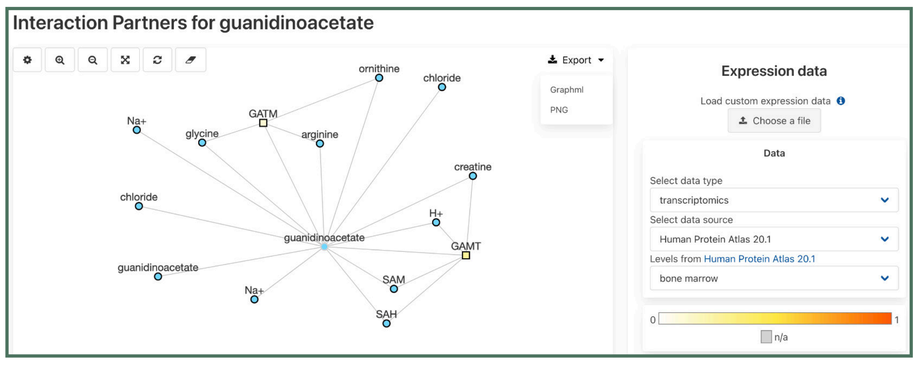
The **history of Metabolic Atlas** began with a focus on the human metabolic model (Pornputtapong et al., 2015). The present website has been re-developed from the ground up, following open-source standards, by first integrating *Human1*, an incorporation and extensive curation of the most recent human metabolic models (Robinson et al., 2020), and *Yeast8*, a consensus metabolic model for *S. cerevisiae* (Lu et al., 2019). In **Metabolic Atlas 1**, a wide range of attributes, amongst which reaction equations, metabolite formulas, gene rules and subsystem contents, are presented in tabular and map views. Following a database redesign and the addition of a performant 3D viewer, 5 more models have been integrated in **Metabolic Atlas 2** (Wang et al., 2021).



The **recently released Metabolic Atlas 3** presents an extensive database with enzyme parameter predicted by AI approaches for interactive web exploration and programmatic access. The initial release of this data resource contains predicted turnover numbers of over 25.7 million enzyme-compound pairs across 8099 organisms (article under consideration). Through the use of cross-references, compounds and genes can be linked to the integrated models (below).

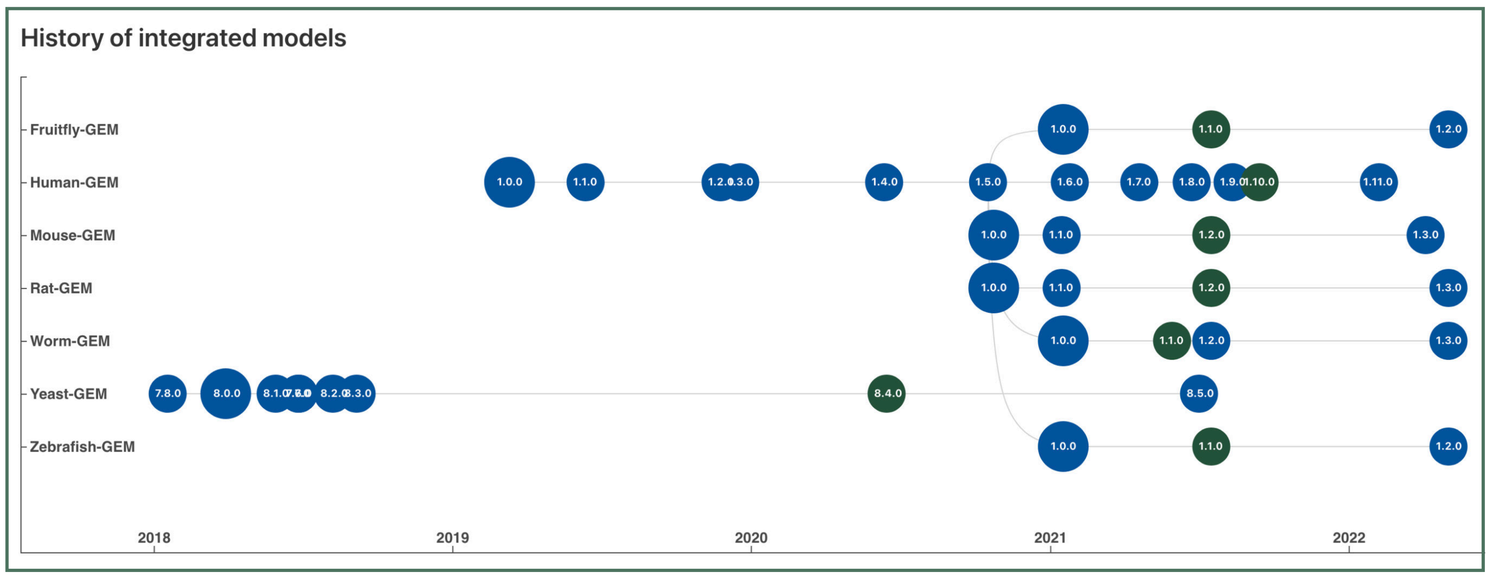
For some of the integrated models, Metabolic Atlas presents manually created **2D maps**. Moreover, it renders **3D maps** for both compartments and subsystems. Both viewers have been extended with a **Data overlay** feature, allowing for graphical interpretation of gene and reaction data (above).





Through the **Interaction Partners,** a dynamically-generated and customisable graph of associations of genes and metabolites, one can interact with a part of the metabolic network. The graph can be further expanded by loading more interaction partners for any element already on the graph. Moreover, RNA expression data from the Human Protein Atlas can be overlaid onto the graph (above).

A new feature of the **GEM repository** has been the addition of an automatically generated figure showing the major and minor updates of the 7 open-source GEMs that Metabolic Atlas integrates (right). Moreover, the GEM repository contains over 350 genome scale metabolic models.



Lu, H., et al, 2019. *A consensus S. cerevisiae metabolic model Yeast8 and its ecosystem for comprehensively probing cellular metabolism*. Nature Commun 10, 3586

Pornputtapong, N., et al, 2015. *Human metabolic atlas: an online resource for human metabolism*. Database, bav068

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